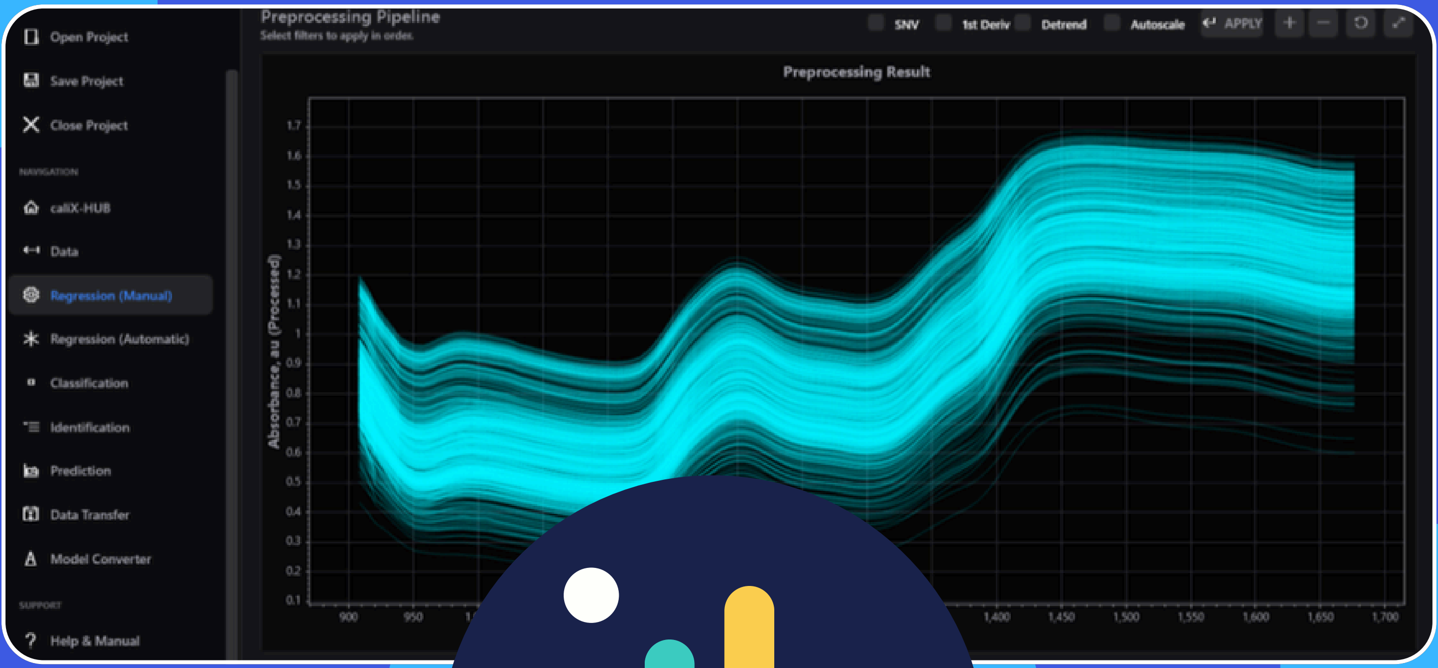
USTECH
INNOVATIONS

caliX

Spectral Intelligence System

2026


Open Project
Save Project
Close Project
NAVIGATION
caliX-HUB
Data
Regression (Manual)
Regression (Automatic)
Classification
Identification
Prediction
Data Transfer
Model Converter
SUPPORT
Help & Manual
Preprocessing Pipeline
Select filters to apply in order.
SNV
1st Deriv
Detrend
Autoscale
APPLY
Preprocessing Result
Absorbance, au (Processed)
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
900
950
1,400
1,450
1,500
1,550
1,600
1,650
1,700

Welcome to USTECH Innovations LLC, a forward-thinking company dedicated to delivering innovative spectral intelligence solutions across industries. We combine cutting-edge AutoML technology, expert talent, and strategic vision to help laboratories and enterprises turn raw spectral data into reliable chemical predictions instantly, allowing businesses to grow and thrive in a digital-first world.

INNOVATING THE FUTURE, DELIVERING EXCELLENCE

Understanding Light, Predicting Chemistry.

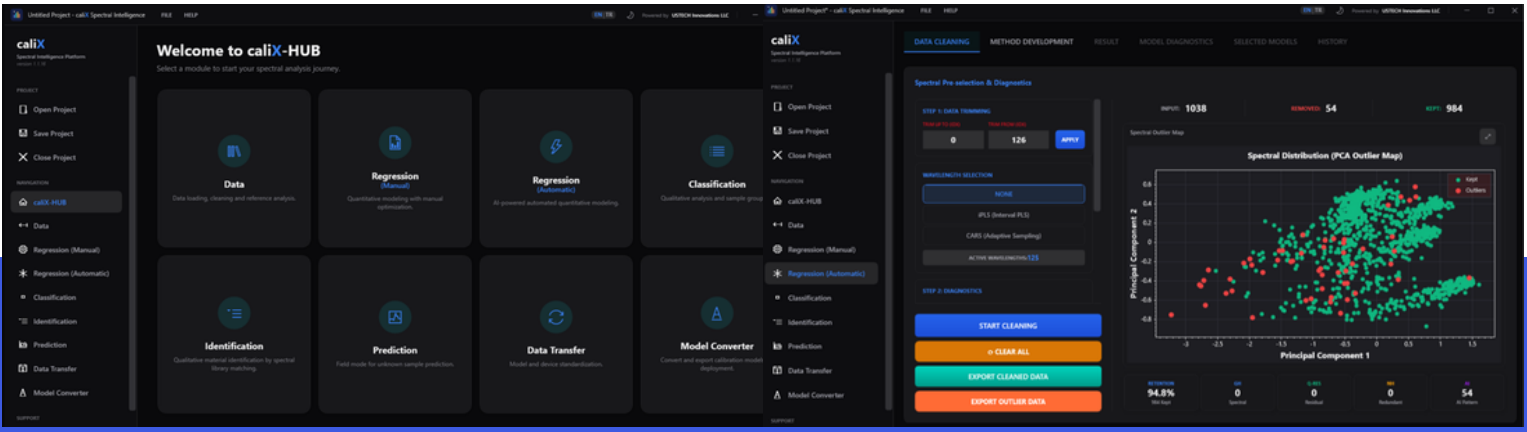
SOFTWARE
FEATURES & BENEFITS

KEY FEATURE

A powerful, intuitive application designed to streamline your workflow and maximize productivity at every level.

HIGHLIGHTS

- Lightning-fast modeling performance and computing times
- One-click professional reporting and automated Excel data export
- Role-based access control and adaptive expert workflows
- Seamless third-party chemometric app integrations (Solo, PLS_Toolbox XML)



Welcome to caliX-HUB

Select a module to start your spectral analysis journey.

Data: Data loading, cleaning and reference analysis.

Regression (Manual): Quantitative modeling with manual optimization.

Regression (Automatic): AI-powered automated quantitative modeling.

Prediction: Field mode for unknown sample prediction.

Data Transfer: Model and device standardization.

Untitled Project - caliX Spectral Intelligence
FILE
HELP
caliX
Spectral Intelligence Platform
Open Project
Save Project
Close Project
caliX-HUB
Data
Regression (Manual)
Regression (Automatic)
Classification
Identification
Prediction
Data Transfer
Model Converter
DATA CLEANING
METHOD DEVELOPMENT
RESULT
MODEL DIAGNOSTICS
SELECTED MODELS
HISTORY
Spectral Pre-selection & Diagnostics
STEP 1: DATA TRIMMING
0
126
APPLY
WAVELENGTH SELECTION
NONE
iPLS (Interval PLS)
CARS (Adaptive Sampling)
ACTIVE WAVELENGTHS: 125
STEP 2: DIAGNOSTICS
START CLEANING
CLEAR ALL
EXPORT CLEANED DATA
EXPORT OUTLIER DATA
INPUT: 1038
REMOVED: 54
KEPT: 984
Spectral Outlier Map
Spectral Distribution (PCA Outlier Map)
Kept
Outliers
Principal Component 2
Principal Component 1
94.8%
0
0
0
54

SYSTEM FEATURES

Explore the full range of features our application offers:

Smart Automation

Evaluates dozens of preprocessing paths, tests variations automatically, and ranks the best models based on cross-validation accuracy.

Stunning UI/UX Graphics

Interactive "Predicted vs. Reference" scatter charts and component optimization grids engineered for a premium tracking experience.

Advanced Spectral Cleaning

High-performance filters (SNV, MSC, 1st/2nd Derivatives) that mathematically strip light scattering and baseline shifts.

Prediction Diagnostics

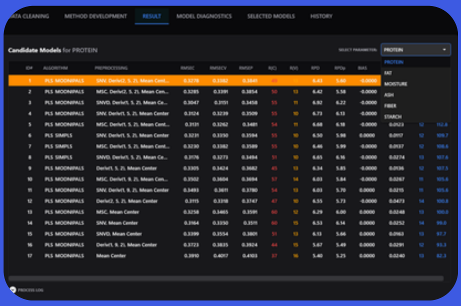
Advanced tracking via Global Hotelling (GH) and Neighbor Hotelling (NH) indicators to red-flag out-of-range samples instantly.

CORE CAPABILITIES

caliX delivers powerful tools and seamless experiences that help teams collaborate, automate, and interpret spectral data efficiently.

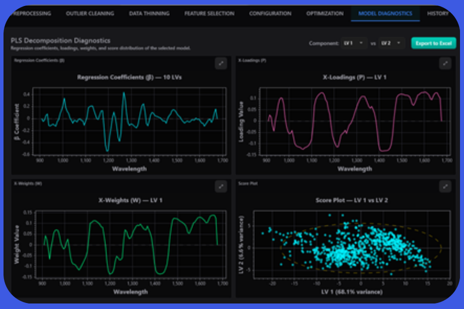
Full Transparency

Complete transparency in data usage, model validation parameters (R², RMSEC, RMSECV), and system metrics.


METHOD DEVELOPMENT
RESULT
MODEL DIAGNOSTICS
SELECTED MODELS
HISTORY
Candidate Models for PROTEIN
PROTEIN
FAT
MOISTURE
ASH
FIBER
STARCH
PROCESS LOG

Continuous Updates

Continuously updated with new chemometric features based on user feedback and evolving spectroscopy needs.

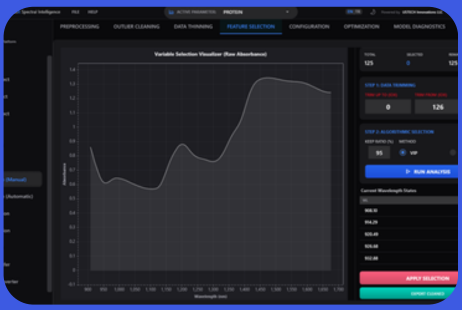

OUTLIER CLEANING
DATA THINNING
FEATURE SELECTION
CONFIGURATION
OPTIMIZATION
MODEL DIAGNOSTICS
HISTORY
PLS Decomposition Diagnostics
Component: LV 1 vs LV 2
Export to Excel
Regression Coefficients (β) — 10 LVs
β Coefficient
Wavelength
X-Loadings (P) — LV 1
Loading Value
Wavelength
X-Weights (W) — LV 1
Weight Value
Wavelength
Score Plot — LV 1 vs LV 2
LV 2 (6.6% variance)
LV 1 (68.1% variance)

Dedicated Support

Dedicated onboarding engineering specialists and rapid technical support within 24 hours.

Team Collaboration

Genuinely shared project files (.clx) allow laboratory teams to sync and transfer calibration data anywhere in real time.


Spectral Intelligence
FILE
HELP
PROTEIN
PREPROCESSING
OUTLIER CLEANING
DATA THINNING
FEATURE SELECTION
CONFIGURATION
OPTIMIZATION
MODEL DIAGNOSTICS
Variable Selection Visualizer (Raw Absorbance)
Absorbance
Wavelength (nm)
125
0
126
95
VIP
RUN ANALYSIS
Current Wavelength States
APPLY SELECTION

User-Centric Design

Every mathematical model and option is designed around user needs to deliver maximum industrial value.

WHY USE OUR SPECTRAL INTELLIGENCE SYSTEM?

Our system is built for enterprise performance and ease of use. Here is what makes it stand out:

- Cross-Instrument Support: Completely hardware-agnostic; seamlessly processes spectral formats from any NIR, FTIR, Raman, or UV-Vis spectrometer.
- Real-Time Synchronicity: All your calibration models are instantly saved, exported, or packed into native .clx or standalone .ihb files.
- Advanced Validation & Security: Built-in Robust PLS and Huber regression algorithms aggressively downweight outliers and prevent model overfitting.
- Custom Dashboards: Personalize your workspace with a simple toggle between professional Light and Dark themes to match your preferred workflow.
- Offline Mode: Continue optimizing algorithms, processing datasets, and predicting unknown samples without an active internet connection.

OUR TEAM

Our team consists of passionate professionals with expertise in software development, data science, and chemometrics, delivering exceptional analytical and digital experiences.

Team Strengths:
- Chemometrics & Model Developers
- UX/UI Interface Designers
- Product & Analytics Strategists
- Agile Collaboration

OUR SYSTEM

Our system is built to streamline workflows and enhance productivity. It combines an intuitive user interface with powerful mathematical engines to serve laboratories and production lines across various industries.

App Features:
- Intuitive Interface
- One-Click Optimization (AutoML)
- Real-Time Spectral Analytics
- Secure & Scalable Platform

MEASURABLE IMPACT

Optimizing calibration models for the instant prediction of Protein, Moisture, Fat, and other critical parameters across automated production lines.

Streamlining complex spectral data from NIR, FTIR, and Raman spectrometers with absolute mathematical precision and zero hardware restrictions.

95%
User & Model Accuracy Satisfaction

90%
Calibration & Analysis Time Savings

85%
Operational Productivity Boost

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Let's build something extraordinary together.